Problem Set III, Macroeconomics III
University of Copenhagen

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Exercise 1 – Government debt (Exam exercise from 2024)

Consider a representative household with preferences given by:

$$U = \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma},$$

for $\sigma > 0$, where $0 < \beta < 1$ is the discount factor, and c_t is per capita consumption. The household supplies labor inelastically ($l_t = 1$), receives income w_t, pays a lump-sum tax T_t, and saves by investing in assets a_t that earn interest r_t. The economy is closed. The budget constraint is:

$$a_{t+1} + c_t = (1 + r_t)a_t + w_t - T_t.$$

a) Write down the Lagrangian for the household's optimization problem and derive the first-order conditions with respect to c_t, a_{t+1}, and the Lagrange multiplier.

b) Show that the household's optimal consumption is described by the Euler equation:

$$\frac{c_{t+1}}{c_t} = \left[\beta\left(1 + r_{t+1}\right)\right]^{1/\sigma}.$$

Explain the economic intuition behind this equation. What does it imply for the relationship between the household's savings decision and the interest rate?

Beyond the first order conditions derived above, the household is subject to a no-Ponzi game condition and a transversality condition. Hence, it can be shown (you don't need to) that:

$$\sum_{t=0}^{\infty} \frac{c_t}{\prod_{s=0}^{t}(1+r_s)} = \sum_{t=0}^{\infty} \frac{w_t - T_t}{\prod_{s=0}^{t}(1+r_s)} + a_0. \tag{1}$$

The representative firm in the economy only uses labor in production. Its production function is described by:

$$y_t = A_t L_t,$$

where $A_t > 0$ is labor productivity, and L_t is the amount of labor hired. The firm maximizes its profits, which are given by:

$$\Pi_t = A_t L_t - w_t L_t,$$

where w_t is the wage rate.

c) Maximize the firm's profits with respect to L_t and derive the equilibrium wage w_t. What happens to wages if A_t increases? Explain the economic intuition. *(Hint: Remember that labor is supplied inelastically by households.)*

Finally, the economy's government has the following budget constraint:

$$B_{t+1} = (1 + r_t)B_t + G_t - T_t \qquad \text{(Gov budget constraint)}$$

Just like the household, the government is subject to a no-ponzi game and a transversality condition, meaning debt cannot grow indefinitely:

$$\lim_{T \to \infty} \frac{B_T}{\prod_{s=0}^{T}(1 + r_s)} = 0. \qquad \text{(Gov debt constraint)}$$

Assume, for now, that the government does not spend any resources $(G_t = 0)$ and has no debt $(B_t = B_{t+1} = 0)$.

d) Using the household's and the government's budget constraints, derive the resource constraint for the economy. In this economy, what is the household's marginal propensity to consume, i.e., $\frac{\partial c_t}{\partial w_t}$? Explain. *(Hint: Remember that in general equilibrium, $B_t = a_t$ for all t.)*

e) Assuming the economy is in steady state, with $A = 1$, derive the equilibrium interest rate r. In an economy with a higher steady state level of productivity $A > 1$, is the interest rate different? Explain intuitively.

Now assume that instead of the previous economy, the representative household lives in a world where government debt is positive, meaning it can save $B_t = a_t > 0$ over time. Also assume that this economy is in steady state in period 0. In this new economy,

in period 0, the government decides, surprisingly, to spend a constant amount G each period, forever – this will be the new steady state. Households are given a choice: pay the discounted value of all spending in period 1, or get taxed $T = G$ each period from now until forever.

f) Show mathematically that households are indifferent between the two tax regimes. Explain intuitively why this is the case. (*Hint: it may be useful to forward-substitute the government budget constraint, and use the government's debt constraint.*)

g) Describe how consumption c_t and the interest rate r_t evolve along the transition to the new steady state? Explain.

Exercise 2 – Credit constraints and Ricardian Equivalence

Suppose a household solves the following variant of the two-period consumption-savings problem:

$$\begin{aligned} &\max_{c_0,a,c_1} u(c_0) + \beta u(c_1) \\ &\text{s.t. } a = y_0 - \tau_0 - c_0 \\ &\quad c_1 = y_1 - \tau_1 + (1+r)a \\ &\quad a \geq -b \qquad \text{(Constraint (q))} \end{aligned}$$

(Hint: notice that the last equation is a weak inequality, not an equality, so it may or may not be binding. If it is not binding, the solution will be as in previous exercises of this nature. If it is binding, the solution will be different. It is easiest to solve each case separately and then compare them to know whether the constraint will or will not be binding. Alternatively, these problems can be solved using the Karush-Kuhn-Tucker approach. In this exercise, you are asked to do the former.)

(a) What does the constraint (q) represent; how exactly does it "constrain" the consumer? What does b represent?

(b) Plot the household's budget constraint. In the same graph, plot the constraint in the last equation. For this, assume some values for y_0, r and y_1. Set $\tau_0 = \tau_1 = 0$. How could you tell, just from your graph, if the constraint will be binding?

(c) Solve the problem as follows: set up the Lagrangian using only the utility function

and the combined budget constraint, ignore constraint (q) for now.[1] Take the first order conditions and show that the consumer's decisions is described by an Euler equation of the form

$$u'(c_0) = \beta(1+r)u'(c_1)$$

What does this equation tell you about the growth rate of consumption if r were to rise?

d) Assume that $u(c) = \ln(c)$. Solve for c_0 as a function of the model's parameters (except for b). This is the agent's "unconstrained" choice. Explain. Mark this choice on you graph.

e) Now solve the alternative case where the constraint is binding: $a = -b$. What are the consumer's choices now? And why is it fair to say that consumers are "off their Euler equation", as in,

$$u'(c_0) > \beta(1+r)u'(c_1)?$$

f) Show that, other things being equal, the inequality constraint is more likely to be binding if

i. $y_1 - \tau_1$ is high,

ii. $y_0 - \tau_0$ is low,

iii. b is low.

Interpret each of these conditions.

g) Suppose that the government announces a "stimulus package" of size Δ. This involves lowering τ_0 by Δ and increasing τ_1 by $\Delta(1+r)$ so that the present value of taxes is unchanged. How does c_0 respond to the stimulus package if we start from a situation where constraint (q) is NOT binding? How does c_0 respond to the stimulus package if we start from a situation where constraint q is binding? Explain.

[1] This is the same as assuming that $a > -b$, i.e., the constraint does not bind.